Properties at a glance

Three properties belong to a vector on its own; three exist only between a pair. The split matters because the zero vector behaves differently on either side of it.

06
PROPERTIES

UNCONDITIONAL 2

i Magnitude $\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$

UNCONDITIONAL

REQUIRES intrinsic — belongs to one vector

A single non-negative number: the length of the arrow, independent of where it is drawn. It is zero exactly for the zero vector, which is the property positive definiteness of the dot product records.

iii Dimension $\mathbf{v} \in \mathbb{R}^n \Rightarrow \dim = n$

UNCONDITIONAL

REQUIRES intrinsic — fixed by the space

The component count, decided by the space rather than the vector. No operation on vectors changes it — addition and scaling stay inside $\mathbb{R}^n$, which is what makes it a vector space rather than merely a set.

CONDITIONAL 4

ii Direction $\hat{\mathbf{v}} = \dfrac{\mathbf{v}}{\|\mathbf{v}\|}$

CONDITIONAL

HOLDS WHEN $\mathbf{v} \neq \mathbf{0}$

FAILS WHEN $\mathbf{v} = \mathbf{0}$ — division by zero

The unit vector pointing the same way. The zero vector has no direction at all, and this is not a convention that could have gone otherwise: the definition divides by a magnitude of zero. Every later statement about $\mathbf{0}$ traces back here.

iv Equality $\mathbf{a} = \mathbf{b} \iff a_i = b_i$ for every i

CONDITIONAL

HOLDS WHEN both in the same $\mathbb{R}^n$

All-or-nothing: one differing component breaks it. Vectors in different spaces are not unequal so much as incomparable — the question does not arise.

v Parallelism $\mathbf{a} \parallel \mathbf{b} \iff \mathbf{a} = c\mathbf{b}$ for some scalar c

CONDITIONAL

HOLDS WHEN either vector may be $\mathbf{0}$

COMMON ERROR assuming parallel means same direction — $c < 0$ gives opposite directions

A relation between two vectors, not a property of either. The zero vector is parallel to everything by convention, which is consistent because $\mathbf{0} = 0 \cdot \mathbf{b}$ holds for every $\mathbf{b}$. In $\mathbb{R}^3$ the cross product detects this: parallel vectors have $\mathbf{a} \times \mathbf{b} = \mathbf{0}$.

vi Orthogonality $\mathbf{a} \perp \mathbf{b} \iff \mathbf{a} \cdot \mathbf{b} = 0$

CONDITIONAL

HOLDS WHEN either vector may be $\mathbf{0}$

Also relational. The zero vector is orthogonal to everything, since $\mathbf{0} \cdot \mathbf{b} = 0$ always — so it is simultaneously parallel and orthogonal to every vector, the only one for which both hold. See orthogonality for what this makes possible.

> **THE ZERO VECTOR CASE**
>
> `0 · b = 0 for every b → 0 ⊥ b`
> `0 = 0 · b for every b → 0 ∥ b`

Notice how often the zero vector is the exception. It has no direction, is parallel to everything, and is orthogonal to everything — three conventions that look arbitrary until you see they all follow from $\mathbf{0}$ having magnitude zero and no defined unit vector.